

Rover Houston — Technical Writeup

GDG AI Hack 2026 · MSI Cut the Cord · Team PoliSa

Team PoliSa

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Houston — On-Device Mars Habitat AI on a 18 GB MacBook

Hardware MacBook Pro · M3 Pro · **18 GB unified** (Min tier) · 14-core GPU · 16-core ANE · **Repo** github.com/landigf/gdgaihack-2026 (main) · **Proof** `scripts/airplane-check.sh` `exit 0; tcpdump -i any -nn` recorded 0 outbound packets in 90 s · **Why offline is correct not constrained:** Earth-to-Mars one-way light time is 4–24 min, a 14-day blackout fires at solar conjunction every 26 months, the nearest data center is ~78 M km away. *Cloud SLAs cannot reach the operator — physics picked offline.*

§0 · “Getting Started” five-phase compliance grid

#	Phase	Where	Receipt (measured)
1	Pick a domain — architectural, not thematic	§1	Mars · 4–24 min light time · 14-day blackouts · cloud physically unreachable.
2	Audit hardware □ model ceiling	§2 + Fig 1	M3 Pro · 18 GB · 14-core GPU. 3B beats 7B 57.6 vs 28.7 tok/s (7B is bandwidth-bound). RAM 9.09/18 GB.
3	Ollama / LM Studio set-up	§2 L1 + §5	Ollama gemma3:4b baseline 43.1 tok/s + MLX-LM Qwen2.5-3B-4bit 57.6 tok/s. One-line swap: <code>LLM_BACKEND={ollama,mlx,auto}</code> .
4	Software integration (not a chatbot)	§1 + §4	Tauri folder picker · Reveal in Finder · citation chips open NASA PDFs in Preview · mic capability · hash-routed Mars dashboard in-app · R3F drill-in · voice + Repair persona.
5	Design for the demo (airplane + graceful)	§4	<code>airplane-check.sh</code> <code>exit 0</code> · <code>tcpdump</code> 0 packets / 90 s · <code>IS_TAURI</code> flag in <code>App.tsx</code> swaps Finder for a friendly “open the .app” overlay in a browser tab — no crash.

1 · Architecture — Phase 1 (domain) · Phase 4 (integration)

Shell Tauri 2 (Rust + WebKit) hosting React 18 + Three.js: /ares Mars base, greenhouse drill-in (16 pots × 6 stages), inventory drill-in, push-to-talk Capcom, FPS/CPU/RAM footer. **Sidecar** Python FastAPI on 127.0.0.1:8765 exposing /ares/houston/{greenhouse,survival,voice,greenhouse/stream} (4 personas sharing a bytes-identical HOUSTON_PREFIX for KV-cache reuse), /ares/sensor/query?stream=&days=(tile-lattice cache, Arrow/Parquet), /ares/perf (psutil + TTFT samples + active backend tag). **LLM gen** (swappable via `LLM_BACKEND`): MLX-LM in-process Qwen2.5-3B-Instruct-4bit (default) OR Ollama gemma3:4b

Q4_K_M. **Embeddings** Ollama nomic-embed-text 768d. **ASR** whisper.cpp ggml-base.en (Metal). **TTS** macOS say. **Corpus** 30 NASA PDFs to 1 292 FAISS chunks (Veggie, APH PH-04, NASA-STD-3001).

2 · Five optimization levers, each measured — Phase 2 (hardware ceiling) · Phase 3 (Ollama + MLX)

#	Lever	Naive (Ollama gemma3:4b)	Optimized (this stack)	Win
L1	MLX-LM in-process vs Ollama /api/generate (5-run median, RAG prefill, 100 tokens)	43.1 tok/s · TTFT 2249 ms	57.6 tok/s · TTFT 1668 ms	+34 % decode, -26 % TTFT
L2	Streaming SSE + frontend ReadableStream (token-by-token render)	1820 ms wait for full response	TTFT 1668 ms then live tokens	First word ~25 % sooner
L3	KV-cache reuse on A2A (Greenhouse to Procedure share HOUS- TON_PREFIX)	8124 ms full A2A wall (warm)	4050 ms A2A wall (warm)	2.00x speedup
L4	Tile-lattice time-window cache (50d cold to 20d subset to 70d superset)	2104 ms (3 cold queries)	72 ms (cache + 20d delta)	29.2x speedup , 100 % subset hit
L5	Code-as-action (CodeAct): LLM emits Python in a sandboxed subprocess (-I -B + setrlimit + 30 s timeout) that reads the tile cache + RAG via the houston SDK	2nd LLM round to format	58–438 ms (trivial / sensor / + plot)	quantitative Q answered without a 2nd decode

L4 invalidates per content-hash (PDF mtime + indexer version) — correctness-safe under corpus updates. **Hardware budget under load:** RAM **9.09 / 18 GB (50 %)**, 9.86 GB headroom; sidecar RSS ~2.0 GB; Ollama 30 MB (embeddings only); R3F scene ≥ 45 FPS during decode.

3 · Tech-Opt 30 % figures + Cut-the-Cord compliance — Phase 5

Brief sub-criteria *model selection · quantization · memory · inference speed* are defended row-by-row in §2; reproducible PNGs (light + dark, make_light_figures.py) live under benchmarks/houston/out/: through-put.png, cache_lattice.png, perf_timeline.png (RAM 9.09 / 18 GB, zero swap), a2a_kv_cache.png (2.0x), hardware_util.png (powermetrics).

Zero cloud: MLX-LM + Ollama embeddings + whisper.cpp + macOS say. airplane-check.sh exit 0; tcpdump 0 outbound packets / 90 s; active backend on /ares/perf.llm_backend. **Integration (Phase 4):** Tauri 2 opens cited PDFs in Preview · folder picker · mic capability · say · sidebar tile hash-routes Finder □ Mars dashboard in one window · 4 personas + ASR + TTS + FAISS + Arrow cache + R3F on the same M3 Pro. **Graceful degradation:** App.tsx IS_TAURI flag swaps the file view for a “download the .app” overlay in browser mode — no invoke crash; Mars still works in pure browser. **Reproduce:** ./scripts/setup.sh && ./scripts/download-mars-corpus.sh && LLM_BACKEND=mlx npm run tauri:dev. Bench: make_figures.py. **EdgeXpert forward:** _build_generator() is a one-line swap; ports to the MSI NPU once mlx-lm exposes a Core ML target.